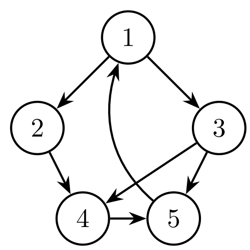
Problem Set 6

Hoa T. Vu

Naive PageRank

Problem 1: Naive PageRank on a Five-Node Graph. Consider the directed graph below with nodes $1, 2, 3, 4, 5$ and edges as follows. The graph is strongly connected and has no dangling nodes.



(a) Write down the naive PageRank equations $\pi_v = \sum_{u \to v} \pi_u / d_u$ for each node v, where d_u is the out-degree of node u.

(b) Together with the normalization constraint $\sum_{v=1}^{5} \pi_v = 1$, solve this system of linear equations by hand to find the naive PageRank vector π. Express each entry as a fraction.

Problem 2: Implementing PageRank. Using the same graph from Problem 1, implement PageRank in Python.

(a) Write out the the transition matrix M.

(b) Starting from the uniform distribution $\pi^{(0)} = [1/5, 1/5, 1/5, 1/5, 1/5]^\top$, perform 10, 100, and 1000 steps of power iteration $\pi^{(t)} = M\, \pi^{(t-1)}$ to compute the probability of ending up at each node. Use NumPy. See the code below.

Code.

```
import numpy as np

nodes = [1, 2, 3, 4, 5]


M = np.array([
    [0, 0, 0, 0, 1],
    [1/2, 0, 0, 0, 0],
```

```python
    [1/2, 0, 0, 0, 0],
    [0, 1, 1/2, 0, 0],
    [0, 0, 1/2, 1, 0],
])

for steps in [10, 100, 1000]:
    pi = np.full(5, 1 / 5)
    for _ in range(steps):
        pi = M @ pi
    print(f"steps = {steps}: {pi}")
```